

2014 JC2 H2 Biology Paper 3 (Suggested mark scheme)

Answer **all** questions.

- 1** A reporter gene is a gene whose activity is relatively easy to detect. Luciferase is an enzyme naturally found in fireflies. When its substrate luciferin is present, it catalyses a chemical reaction which emits light, as shown by the equation below. Therefore, luciferase gene can be used as a reporter gene in recombinant DNA technology.

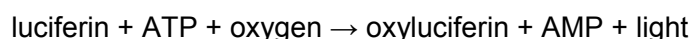


Fig. 1.1 shows a recombinant plasmid construct named “pMCS-Red Firefly Luc” used to study the regulation of *lac* operon in bacteria.

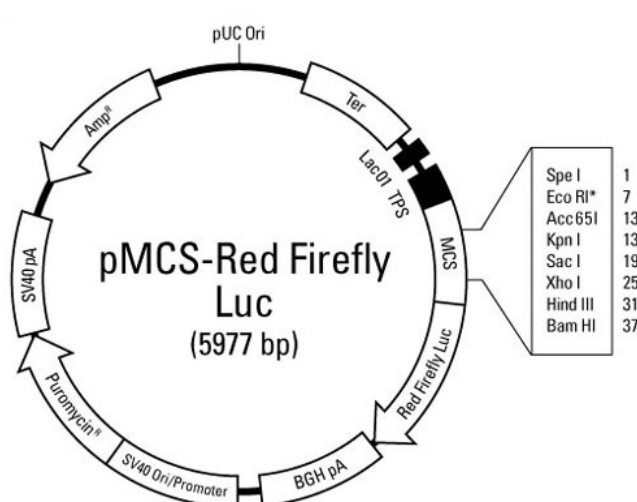


Fig. 1.1

Fig 1.2 shows the detailed structure of Red Firefly Luc insert.

promoter of <i>lac I</i> gene	<i>lac I</i> gene	CAP binding site	promoter	operator	luciferase gene
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Fig. 1.2

- (a)** State the factors which you will need to consider when choosing the right [2]
restriction enzyme to construct the pMCS-Red Firefly Luc.

- (b) Both genomic and cDNA libraries of firefly are available. State where you will take the luciferase gene from for this experiment. Justify your decision. [3]

- (c) Plasmids are introduced into bacteria via electroporation. [2]

Outline how you can select bacterial cells transformed with recombinant plasmid containing the Red Firefly Luc insert.

- (d) A few experiments are conducted to study the regulation of *lac* operon via expression of luciferase gene in the bacterial colonies. [1]

Complete the table below by putting in '+' (low light intensity), '++' (high light intensity) and '-' (no light emitted) in the last column.

<i>experiment</i>	<i>glucose</i>	<i>lactose</i>	<i>ampicillin</i>	<i>luciferin</i>	<i>brightness of colonies</i>
1	✓	x	✓	✓	-
2	✓	✓	✓	✓	+
3	x	✓	✓	✓	++
4	x	✓	✓	x	-
5	x	x	✓	✓	-

- (e) Explain the expected brightness of colonies for experiments 1 and 3. [4]

- (f) Outline the problems of cloning eukaryotic genes in prokaryotic cells. [2]

[Total: 14]

- 2 Cystic fibrosis is a genetic disease caused by mutations in the cystic fibrosis transmembrane regulator (CFTR) gene. An experiment utilising genetic engineering techniques was conducted to obtain genetically modified stem cells from monkeys suffering from cystic fibrosis.

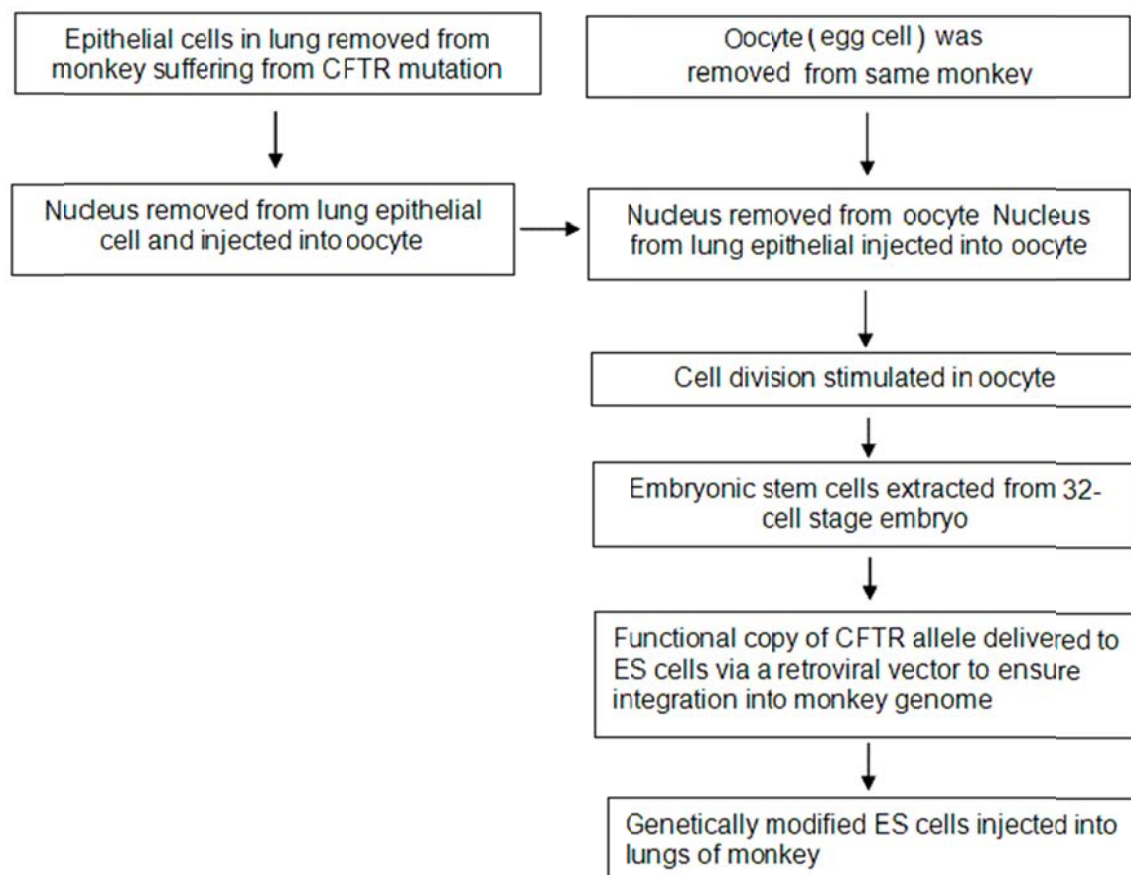


Fig. 2.1

(a) With reference to Fig. 2.1,

- (i) describe the characteristics of the ES cells produced.

[2]

- (ii) suggest why an oocyte was used and why the nucleus had to be [2]
removed from the oocyte.

- (b) In another experiment, modified adenoviral vectors were used to deliver the functional copy of the CFTR allele into the nasal epithelium of monkeys with cystic fibrosis. Eight monkeys with cystic fibrosis were administered with different doses of the modified adenoviral vectors and empty adenoviral vectors. Each monkey was exposed to each type of vector once on the nasal epithelium.

Efficacy of gene transfer was determined six days after the administration of each respective vector. The results of the experiment are shown in Table 2.1.

Table 2.1

Monkey	gender / age	dose / plaque- forming unit (pfu)	presence of functional CFTR mRNA	
			treated with empty adenoviral vector	treated with modified adenoviral vector
1	Female / 25	2×10^7	No	No
2	Male / 44	2×10^7	No	No
3	Male / 23	2×10^8	No	No
4	Male / 21	2×10^8	No	No
5	Female / 34	2×10^9	No	Yes
6	Male / 40	2×10^9	No	No
7	Female / 19	2×10^{10}	No	Yes
8	Female / 19	2×10^{10}	No	Yes

plaque-forming unit (pfu): a measure on the quantity of virus particles

- (i) Describe the main trends shown in Table 2.1. [2]

- (ii) Suggest why the cystic fibrosis monkey was also treated with empty adenoviral vectors. [1]

Intervertebral discs sit between the vertebrae in the spine. They act as shock absorbers and allow movement of the spine. Over time, or following injury or disease, the discs wear out and can lead to back pain, either directly from the discs themselves or because of the pressure on the adjacent spinal cord.

Biomaterials are synthetic materials designed for use in the body. When used in conjunction with stem cells, they can act as a scaffold on which new tissue is formed. Biomaterials seeded with adult stem cells may hold the key to regenerating the damaged disc tissue and reducing pain. Two approaches are currently studied in laboratories. One approach is to produce whole, synthetic intervertebral discs seeded with stem cells. These synthetic discs look like normal discs and the cells grow on them. Alternatively, an artificial, thermosensitive hydrogel seeded with stem cells can be injected into the centre of a synthetic disc to form a fully functioning tissue. Over time, the artificial biomaterial disc and hydrogel biodegrade, leaving natural disc tissue produced by the cells themselves.

- (c) Suggest why adult stem cells, rather than embryonic stem cells, are used [3]
when developing such stem cell-based therapies for the treatment of back
pain.

[Total: 10]

- 3 One of the difficulties of culturing callus is surface sterilizing the piece of plant tissue (explants) to be used, especially when samples are being taken from the wild in tropical countries. In a study, plant tissue samples were taken at three different times of the year and placed on a medium containing no fungicide or antibiotic.

Two different sampling methods were used:

- discs cut from leaves wiped with bleach;
- stem tissues extracted with a sterile hypodermic syringe.

The results of culturing the explants are shown in **Table 3.1**.

Table 3.1

explant	time of year	number of explants	number of cultured explants with no fungal or bacterial contamination
leaf disc	April	153	12
	August	322	16
	January	332	30
stem tissue	April	194	116
	August	191	122
	January	211	156

- (a) (i) Describe how explants may be grown into callus. [3]

- (ii) With reference to **Table 3.1**, compare the effect of the different methods of taking tissue samples and the time of year on fungal and bacterial contamination of the cultured explants. [3]

- (iii) Suggest a possible reason for the differences observed between the two types of explant. [2]

Sulfonylurea which is a major component of a herbicide acts by binding to the enzyme acetolactate synthase (ALS), causing dysfunction of the enzyme and reducing the synthesis of certain amino acids that are necessary for protein synthesis. A mutation in the ALS gene found in weed was responsible for resistance to sulfonylurea.

An experiment was performed to study the effect of inserting the normal and mutated ALS gene from the weed into rice cells. The rice cells were grown in tissue culture and divided into four groups (**A**, **B**, **C** and **D**) according to their treatment as follows:

- A** : not genetically engineered
- B** : genetically engineered to express the normal ALS gene from weed
- C** : genetically engineered to express one copy of mutated ALS gene from weed
- D** : genetically engineered to express two copies of the mutated ALS gene from weed

Each group of cells was then grown into callus.

Fig. 3.1 shows the effect of different concentrations of sulfonylurea herbicide on the growth of each rice callus.

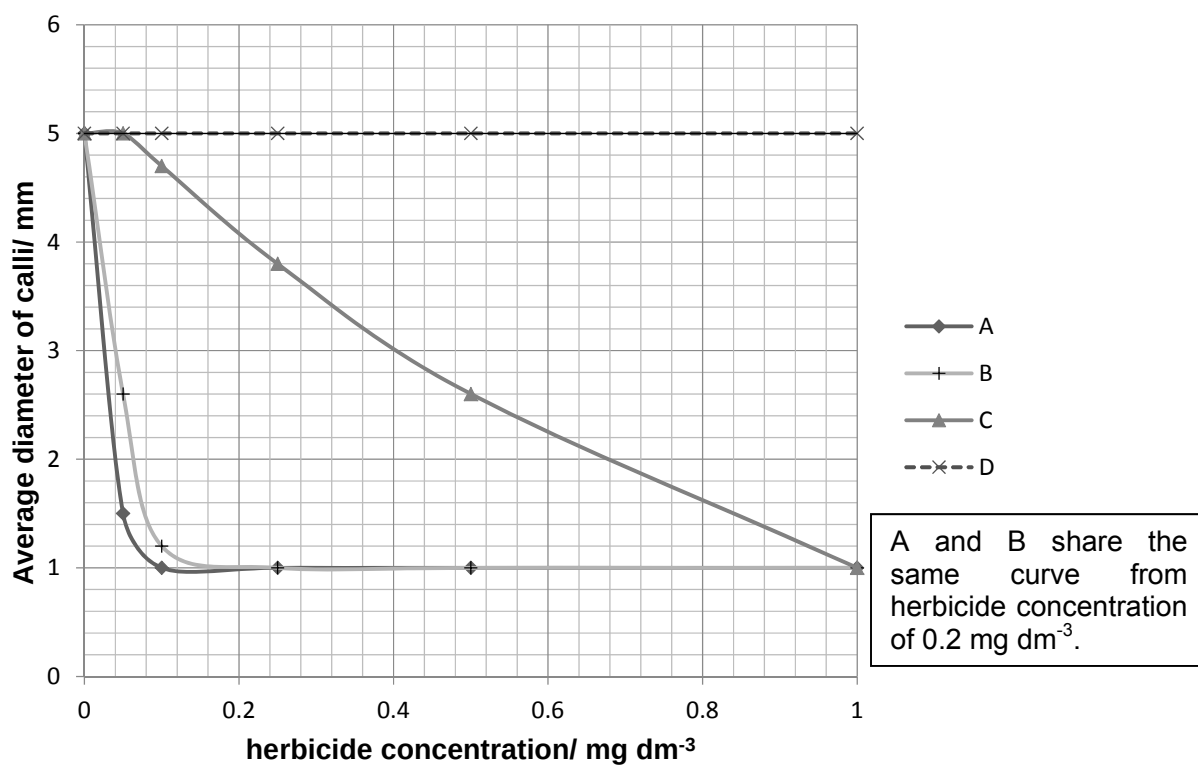


Fig. 3.1

(b) (i) Explain the purpose of calli **A** and **B** in this investigation.

[2]

- (ii) State three advantages of using callus cells in producing genetically modified rice. [3]

- (iii) With reference to **Fig. 3.1**, assess the effect of mutated ALS gene on the expression of sulfonylurea resistance. Justify your answer. [3]

[Total: 16]

4 Planning question

Cell division in flowering plants takes place in particular regions of the plant called meristems. Cells in meristems are not specialized for any particular function and divide repeatedly by mitosis. Some of the daughter cells remain meristematic; others cease dividing and become differentiated into appropriate cell types depending on their position.

Mitotic index is a measure for the proliferation status of a cell population. It is defined as the ratio between the number of cells in mitosis and the total number of cells. This can be done by observing cells undergoing cell division with a microscope.

You are required to plan an investigation on the difference in duration of each stage in the mitotic cycle, with respect to one another, in an actively dividing cell. This is to establish the longest stage in mitosis. In this planning, you will be provided with *Allium sativum*, commonly known as garlic.

Your planning must be based on the assumption that you have been provided with the following equipment and materials which you should use:

- a microscope
- *A. sativum* roots with intact tips
- toluidine blue which stains chromosomes blue
- 1M hydrochloric acid which fixes the specimen
- 70% ethanol
- Thermostatically water bath at 60 °C
- Scissor
- Watch glass (small glass dish)
- Beaker, 100 cm³,
- Scalpel
- Mounted needle
- Microscope slide
- Coverslip

Your plan should:

- include a clear statement of the hypothesis or prediction
- identify the key variables
- give full details and explanations of the procedures that you would adopt to ensure that the results are as precise and reliable as possible
- relevant, clearly labelled diagrams
- show how you would present and analyse your results
- include a brief risk assessment
- be written in clear scientific language, using the correct technical and scientific terms.

[Total: 12]

Free-response question

- 5 (a) RFLP is a method used by molecular biologists to follow a particular [8]
sequence of DNA as it is passed on to other cells. RFLPs can be used in
many different settings to accomplish different objectives.

Explain how RFLP analysis facilitates the process of DNA fingerprinting and
genomic mapping.

- (b) The polymerase chain reaction (PCR) is arguably the most powerful [8]
laboratory technique ever invented.

Describe PCR and explain the advantages and limitations of this technique.

- (c) The Human Genome Project (HGP) was one of the great feats of exploration [4]
in history - an inward voyage of discovery using techniques of RFLP and
PCR.

Describe the goals of the human genome project.

The End